Supplementary Information

Ontology-constrained multi-LLM scoring of hypothesis support in the predictive processing literature

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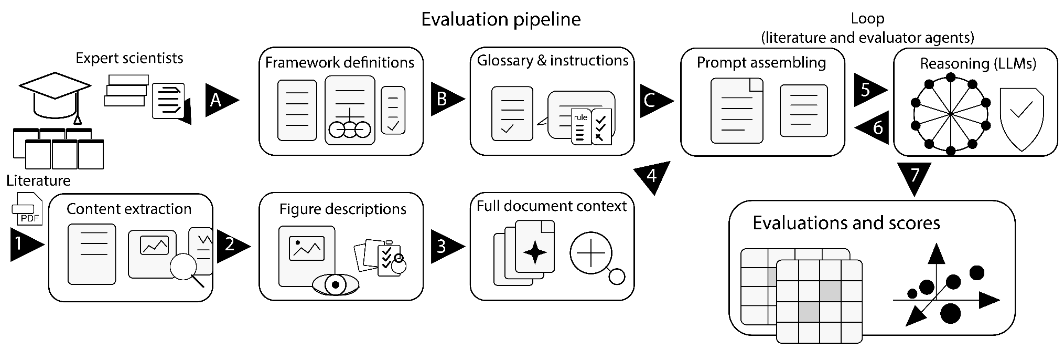
Supplementary Information

Supplementary Figures

Supplementary Fig. S1

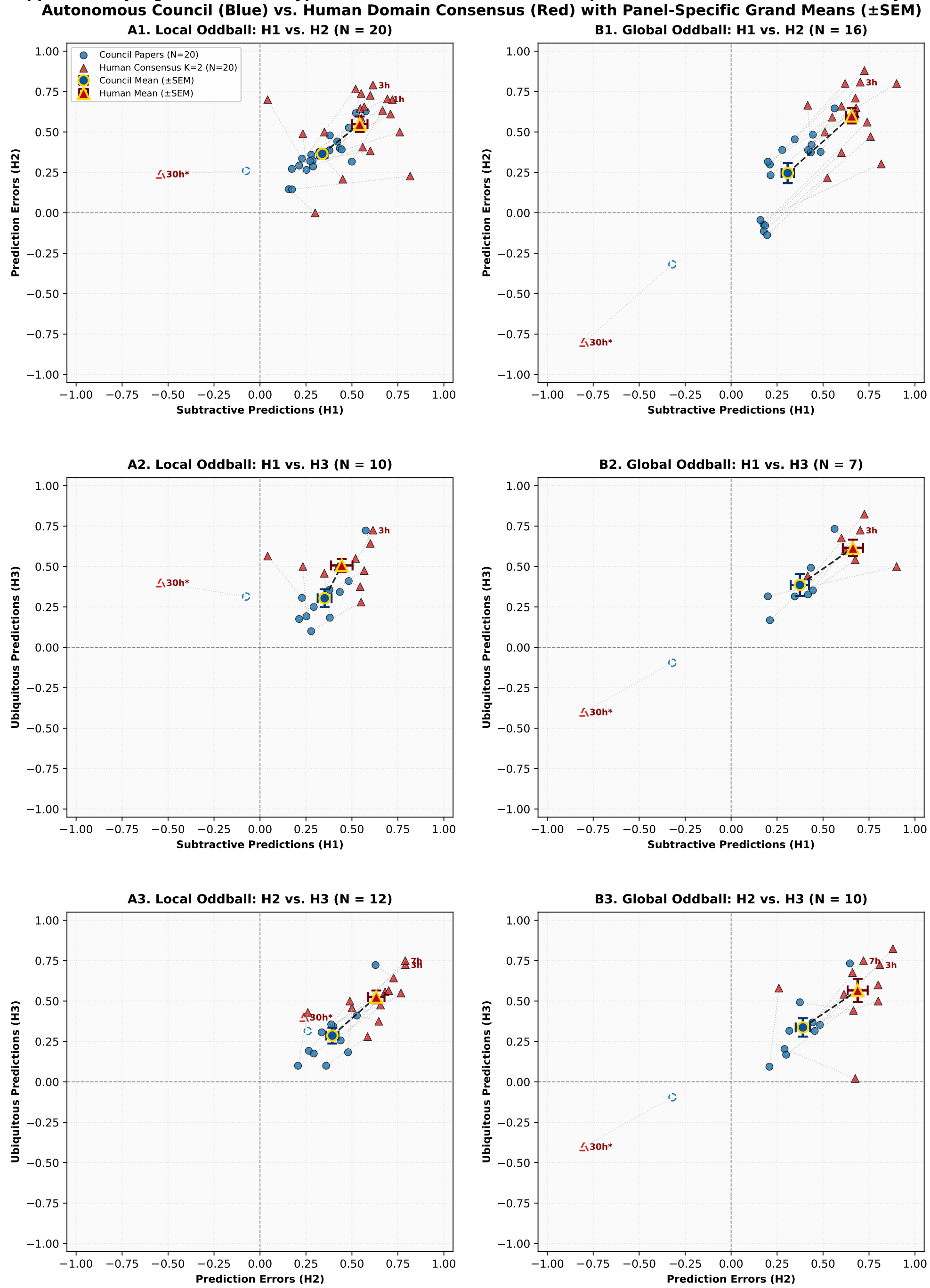
Supplementary Fig. S1: Evaluation Pipeline Architecture. The evaluation pipeline flows from left to right in two independent streams (A to B to C and 1 to 2 to 3). The A to B to C pathway establishes the instructions, canonical glossary, and prompt for the models. The 1 to 2 to 3 pathway ingests the paper and combines figures with text to provide a single text object for the models. Steps 5 and 6 loop between models and papers until all evaluations are completed. The final step 7 quantifies and visualizes the evaluations.



Supplementary Fig. S2

Supplementary Fig. S2: ~~**Independent Human Expert Concordance (**$K = 2$ **domain experts) . (A)Factor-cell level concordance between human consensus (**$N = 175$ **non-null paired factor cells)**~~ **Pairwise 2D Hypothesis-Space Projections** **and** ~~**the autonomous LLM council consensus (**$r = 0.162, \rho = 0.241,$ **MAE** $= 0.301,$ **Directional Concordanc . (B)Study-level aggregation across** $N = 25$ **common evaluable papers(Pearson** $r = 0.529,$ **Spearman** $\rho = 0.492$**) , with stronger study-level than factor-cell concordance**~~**Human– Concordance.** Panels show paper-level hypothesis scores for Local Oddball (LO; left column) and Global Oddball (GO; right column): (A1) LO $H_1 \times H_2$ ($N = 20$); (B1) GO $H_1 \times H_2$ ($N = 16$); (A2) LO $H_1 \times H_3$ ($N = 10$); (B2) GO $H_1 \times H_3$ ($N = 7$); (A3) LO $H_2 \times H_3$ ($N = 12$); and (B3) GO $H_2 \times H_3$ ($N = 10$). Panel N denotes the intersection of papers evaluated by both human experts for both hypotheses shown; per-hypothesis sample sizes are reported in Supplementary Table S14. Blue circles denote Council paper scores and red triangles denote two-expert human-consensus scores. Large gold-bordered markers show panel-specific means ± SEM across the same independent papers. Dotted gray lines connect paired evaluations for the same study. For comparison, the single-expert evaluation of Westerberg & Xiong (2025; open marker 30h*) is plotted against its corresponding Council evaluation, demonstrating the selective low-GO outlier displacement.

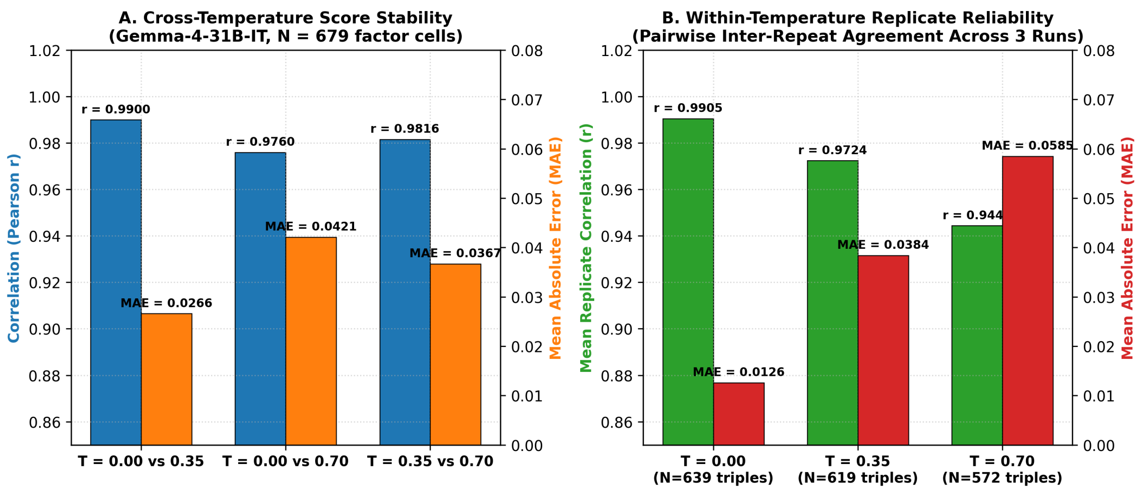
Supplementary Fig. S2: Pairwise Hypothesis Concordance across Independent Literature Studies (3×2 Projection)
Autonomous Council (Blue) vs. Human Domain Consensus (Red) with Panel-Specific Grand Means (±SEM)



Panels display exact K=2 intersection papers evaluated by both domain experts for the two plotted hypotheses (A1: N=20, B1: N=16, A2: N=10, B2: N=7, A3: N=12, B3: N=10).
Large gold-bordered markers show panel-specific Human and Council Grand Means ± across-study SEM (SEM = SD_papers / sqrt(N_panel)). Dotted grey lines connect paired evaluations.
30h*: Westerberg & Xiong (2025) single-expert benchmark (open marker; LO: [-0.54, +0.50, +0.40], GO: [-0.80, -0.80, -0.40]). Landmark K=2 studies: 01h: Attinger2017 | 03h: Bastos2012 | 07h: Friston2010.

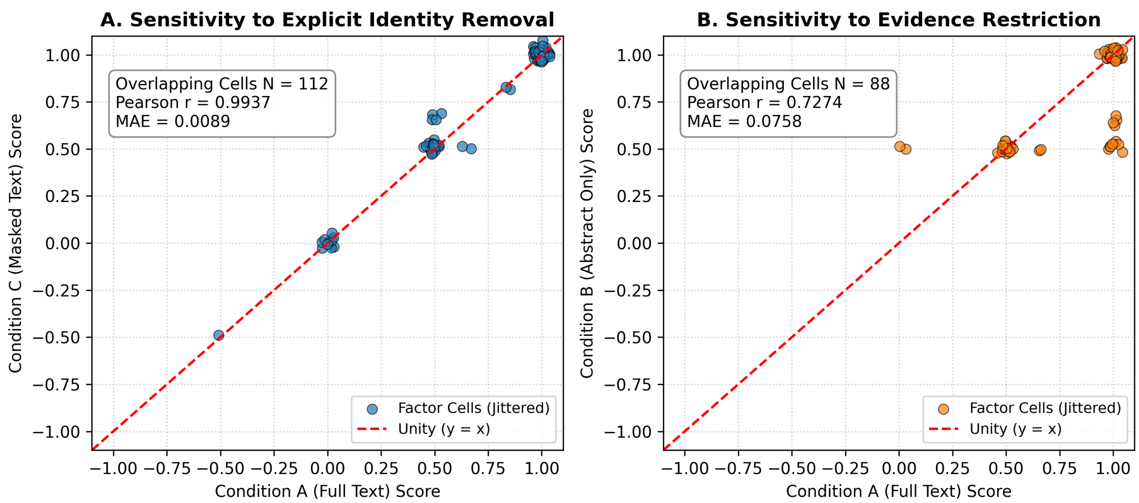
Supplementary Fig. S3

Supplementary Fig. S3: ~~837-Call Factorial Robustness Analysis (31 papers × 3 models × 3 temperatures × 3 repeats)~~Decoding Robustness and Stochastic Replicate Reliability Analysis. (A) ~~Score distribution stability~~ Cross-temperature score stability in the primary complete evaluation model (`gemma-4-31b-it`; 279 calls across 31 papers), showing Pearson correlations ($r \geq 0.9760$) and mean absolute errors (MAE ≤ 0.0421) across sampling temperatures ~~(~~$T \in \{0.00, 0.35, 0.70\}$~~) for OLMo-3-32B-Think, Gemma-4-31B-IT, and Phi-4-Reasoning-Plus~~. (B) ~~Context displacement stability ($\Delta_{\text{LO-GO}}$)demonstrating that the primary scientific contrast remains stable across the tested decoding settings and stochastic repeats~~Replicate reliability across independent stochastic generations held at fixed temperatures ($T = 0.00 : \bar{r} = 0.9905, \text{MAE} = 0.0126$ $T = 0.35 : \bar{r} = 0.9724, \text{MAE} = 0.0384$; $T = 0.70 : \bar{r} = 0.9444, \text{MAE} = 0.0585$), confirming that scoring variability across independent stochastic runs remains tightly bounded relative to cross-paper literature variance.



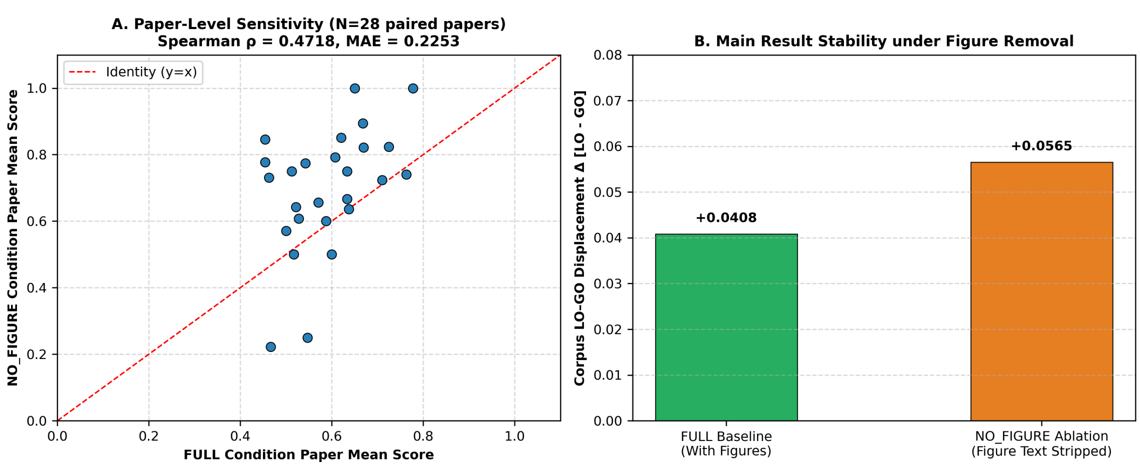
Supplementary Fig. S4

Supplementary Fig. S4: Evidence ~~Grounding vs. Prior Bias Sensitivity Analysis(54 inference calls across intact, abstract-only, and masked-manuscript evidence controls). (A) Sensitivity to explicit study/author identity removal (Condition AFull Text vs Condition C Masked Text: Pearson $r = 0.9937$, MAE = 0.0089~~ Availability and Publication-Identity Sensitivity Analysis. Evaluates factor-cell score correspondence between intact manuscript text (FULL) and de-identified text (MASKED, Panel A; Pearson $r = 0.9937$, MAE $= 0.0089$ across $N = 112$ overlapping factor cells) ~~, showing minimal sensitivity to explicit study/author identity removal. (B) Sensitivity to evidence restriction (Condition A Full Text vs Condition B Abstract Only: Pearson $r = 0.7274$, MAE = 0.0758~~ and between intact manuscript text and restricted text (ABSTRACT Only, Panel B; Pearson $r = 0.7274$, MAE $= 0.0758$ across $N = 88$ overlapping factor cells), ~~showing score changes when detailed empirical text is withheld. Full, Shuffled, and Prior-only condition summaries are reported in Table S10.~~ demonstrating that factor scoring is robust to publication-identity removal but sensitive to restriction of available empirical evidence.



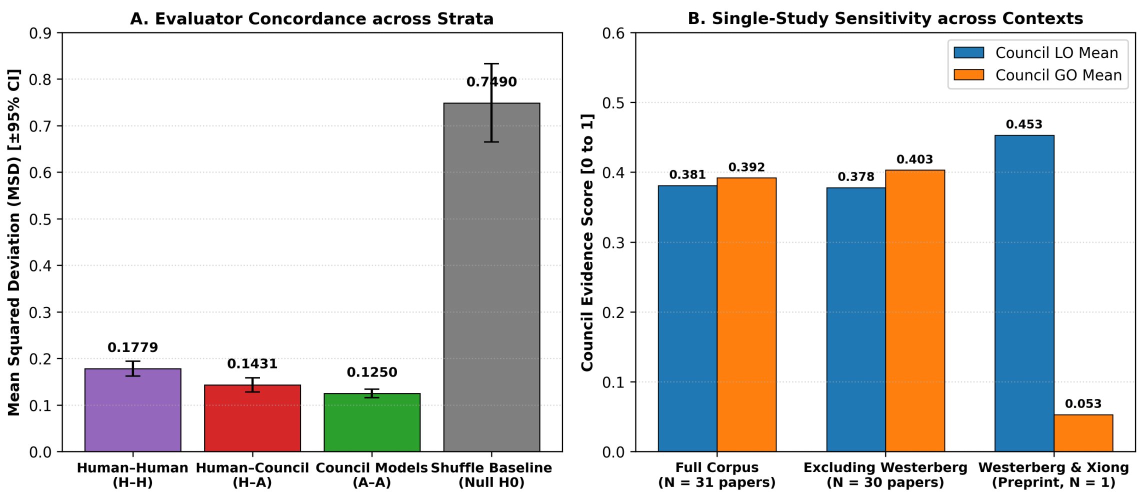
Supplementary Fig. S5

Supplementary Fig. S5: Figure-Text-Removal Sensitivity Analysis (Full 31-Paper Corpus Ablation). (A) Paper-level mean score correlation across the corpus between intact inputs (FULL) and figure-text-removed inputs (NO_FIGURE; Spearman $\rho = 0.4718$ ~~,~~ across $N = 28$ paired empirical paper means; factor-cell level MAE = 0.2253 across all 533 paired ~~factor~~ cells nested in 31 papers). (B) ~~Corpus-level LO − GO context displacement stability across 28 paired empirical papers ($\Delta_{\text{Full}} = +0.0408$ vs $\Delta_{\text{NoFig}} = +0.0565$;~~ Factor-cell level directional concordance and stability across the corpus (directional concordance = 99.42% on non-zero cells [518/521], all-cell sign agreement = 97.19% [518/533]~~), showing that the corpus-level contrast retained its direction despite~~, Pearson $r = 0.8539$), demonstrating that while removing figure descriptions produces measurable paper-level score ~~changes~~adjustments, overall factor directionality was preserved across the literature corpus, while paper-level rank correspondence was moderate.



Supplementary Fig. S6

Supplementary Fig. S6: Evaluator ~~Strata Mean Squared Deviation~~ Concordance Strata and Single-Study ~~Westerberg Sensitivity~~ Sensitivity Analysis. (A) ~~Evaluator strata mean squared deviation (MSD $\pm$ 95% CI) across human–human inter-rater reliability (H–$H = 0.1779$), human consensus vs. autonomous~~ Mean squared difference (MSD) across evaluator strata ($K = 2$ human domain experts and autonomous LLM council) compared against a null permutation reference distribution (H_0, $B = 10{,}000$ iterations). Agreement within human experts (H–$H = 0.1779$) and between human consensus and council (H–A = 0.1431) ~~, council inter-model agreement (A–$A = 0.1250$), and the permutation shuffle baseline (Null $H_0 = 0.7490$~~is significantly tighter than chance ($H_0 = 0.7490$, $p_{\text{perm}} = 0.0001$, 0/10,000 extreme permutations).
(B) ~~Single-study sensitivity of corpus displacement ($\Delta_{\text{LO-GO}} \pm$ 95% CI) across the full paired corpus ($N = 28, \Delta = -0.0177$~~Council condition means for Local Oddball (LO) and Global Oddball (GO) across the entire corpus ($N = 31$), excluding Westerberg & Xiong (2025~~)~~ ~~($N = 27, \Delta = +0.0297$)~~; $N = 30$), and for Westerberg & Xiong alone (~~2025) alone ($N = 1, \Delta = -1.2985$)~~N = demonstrating that the aggregate corpus contrast is robust across published studies and that the preprint represents a selective low-GO outlier.



Supplementary Tables

~~**Factor Number Factor Name Definition Context Scope** 1 Subtractive Inhibition (SST) Linear input suppression via somatic inhibition (Somatostatin-mediated). LO+GO 2 Divisive Inhibition (PV) Multiplicative gain reduction (Parvalbumin-mediated). LO+GO 3 Inhibition (GABA) Chloride-mediated inhibition for prediction suppression. LO+GO 4 Habituation to Sequence Habituation suppresses local oddball response relative to a novel local oddball. LO 5 Synaptic Depression (Adaptation) Passive fatigue of synaptic efficacy due to repetition of stimulus. LO 6 Activity Suppression Reduction in firing rates for expected (predictable) stimuli. LO+GO 7 Selective Sharpening Signal-to-noise increase for predictable stimuli, where noise is selectively suppressed to highlight relevant information. LO+GO 8 Alpha/Beta Mediated Suppression Low-frequency bands (8-30Hz) associated with predictions, inhibiting prediction error signals. LO+GO 9 VIP-Mediated Disinhibition VIP-mediated inhibition of SST/PV neurons, leading to disinhibited pyramidal activity during prediction error. LO+GO 10 Precision Weighting (Gain) Top-down amplification of error units, leading to disinhibited gain in response to attended or highly precise stimuli. LO+GO 11 E/I Balance Shift Dynamic adjustment toward higher inhibition for predictable stimuli, leading to suppressed neural firing. LO+GO 12 Omission Response A generated internal signal due to absence of expected input LO+GO~~

#	Factor Name	Definition
1	Subtractive Inhibition (SST)	Linear input suppressi (Somatostatin-mediate
2	Divisive Inhibition (PV)	Multiplicative gain rec
3	Inhibition (GABA)	Chloride-mediated inh
4	Habituation to Sequence	Habituation suppresse to a novel local oddba
5	Synaptic Depression (Adaptation)	Passive fatigue of syna stimulus.
6	Activity Suppression	Reduction in firing rat stimuli.
7	Selective Sharpening	Signal-to-noise increas noise is selectively sup information.
8	Alpha/Beta Mediated Suppression	Low-frequency bands predictions, inhibiting
9	VIP-Mediated Disinhibition	VIP-mediated inhibitio disinhibited pyramidal
10	Precision Weighting (Gain)	Top-down amplificatio disinhibited gain in re precise stimuli.
11	E/I Balance Shift	Dynamic adjustment t predictable stimuli, lea
12	Omission Response	A generated internal s input.

Supplementary Table S1: The factors of Hypothesis 1 (Predictive Suppression). LO =

local oddball. GO = global oddball.

~~**Factor Number Factor Name Definition Context Scope** 13 Error Unit Depolarization Supragranular pyramidal excitation driving ascending error signals. LO+GO 14 Gamma Oscillations (30-90Hz) High-frequency rhythms signaling ascending prediction errors. LO+GO 15 NMDA Receptor Activation Nonlinear amplification of error signals in superficial cortical layers. LO+GO 16 Feedforward Laminar Flow Supragranular-to-granular/supragranular ascending anatomical projection of errors. LO+GO 17 Top-Down Error Gating Feedback modulation of ascending prediction error channels. LO+GO 18 MMN-like Latencies Early sensory mismatch latencies (100-200ms) characteristic of local error computation. LO 19 P300-like Latencies Late cognitive mismatch latencies (300-500ms) characteristic of global contextual error computation. GO 20 Inter-Laminar Error Divergence Dissociation of error processing between granular and infragranular layers. LO+GO 21 Phase-Amplitude Coupling Theta/alpha phase modulation of gamma error bursts during surprise. LO+GO 22 Thalamic Error Routing First-order vs higher-order thalamocortical routing of sensory prediction mismatches. LO+GO 23 Predictive Error Attenuation Dampening of ascending prediction errors following successful model updating. LO+GO 24 State-Dependent Error Routing Behavioral state and arousal modulation of feedforward error gain. LO+GO~~

#	**Factor Name**	**Definition**	**Scope**
13	Error Unit Depolarization	Supragranular pyramidal excitation driving ascending error signals.	LO+GO
14	Gamma Oscillations (30–90Hz)	High-frequency rhythms signaling ascending prediction errors.	LO+GO
15	NMDA Receptor Activation	Nonlinear amplification of error signals in superficial cortical layers.	LO+GO
16	Feedforward Laminar Flow	Supragranular-to-granular/supragranular ascending anatomical projection of errors.	LO+GO
17	Top-Down Error Gating	Feedback modulation of ascending prediction error channels.	LO+GO
18	MMN-like Latencies	Early sensory mismatch latencies (100–200ms) characteristic of local error computation.	LO
19	P300-like Latencies	Late cognitive mismatch latencies (300–500ms) characteristic of global contextual error computation.	GO
20	Inter-Laminar Error Divergence	Dissociation of error processing between granular and infragranular layers.	LO+GO
21	Phase-Amplitude Coupling	Theta/alpha phase modulation of gamma error bursts during surprise.	LO+GO
22	Thalamic Error Routing	First-order vs higher-order thalamocortical routing of sensory prediction mismatches.	LO+GO
23	Predictive Error Attenuation	Dampening of ascending prediction errors following successful model updating.	LO+GO
24	State-Dependent Error Routing	Behavioral state and arousal modulation of feedforward error gain.	LO+GO

Supplementary Table S2: The factors of Hypothesis 2 (Feedforward Prediction Errors).

~~**Factor Number**~~	~~**Factor Name**~~	~~**Definition**~~	~~**Context Scope**~~
~~25~~	~~Cross-Modal Ubiquity~~	~~Implementation of predictive coding mechanisms across multiple sensory modalities (visual, auditory, somatosensory).~~	~~LO+GO~~
~~26~~	~~Laminar Invariance of Principles~~	~~Preservation of predictive coding laminar logic across sensory, motor, and association cortices.~~	~~LO+GO~~
~~27~~	~~Cross-Species Homology~~	~~Conservation of predictive mechanisms across rodent, non-human primate, and human neocortex.~~	~~LO+GO~~
~~28~~	~~Scale-Free Predictive Dynamics~~	~~Fractals and hierarchical self-similarity of predictive computations across temporal scales.~~	~~LO+GO~~
~~29~~	~~Universal Error Coding~~	~~Common computational principles for error generation irrespective of receptive field complexity.~~	~~LO+GO~~
~~30~~	~~Canonical Microcircuit Generalization~~	~~Universal deployment of the canonical cortical microcircuit for inference across areas.~~	~~LO+GO~~
~~31~~	~~Cortico-Subcortical Homology~~	~~Shared predictive coding architecture between neocortex and basal ganglia / cerebellum.~~	~~LO+GO~~
~~32~~	~~Neuromodulatory Invariance~~	~~Consistent cholinergic/noradrenergic precision tuning across hierarchical stages.~~	~~LO+GO~~
~~33~~	~~Temporal Stability of Ubiquity~~	~~The consistent and non-transient presence of these mechanisms across the hierarchy over long recording sessions.~~	~~LO+GO~~
~~34~~	~~Hierarchical Order Stability~~	~~Evidence that the predictive mechanism is ubiquitous and not localized to a single hierarchical pole.~~	~~LO+GO~~
~~35~~	~~Population-Wide Ubiquity~~	~~Consistency of the mechanism across diverse neural populations and cell types throughout the hierarchy.~~	~~LO+GO~~
~~36~~	~~State-Independent Ubiquity~~	~~Presence of the mechanism across different brain states (e.g., wakefulness, sleep, anesthesia) throughout the hierarchy.~~	~~LO+GO~~

#	Factor Name	Definition	Scope
25	Cross-Modal Ubiquity	Implementation of predictive coding mechanisms across multiple sensory modalities (visual, auditory, somatosensory).	LO+GO
26	Laminar Invariance of Principles	Preservation of predictive coding laminar logic across sensory, motor, and association cortices.	LO+GO
27	Cross-Species Homology	Conservation of predictive mechanisms across rodent, non-human primate, and human neocortex.	LO+GO
28	Scale-Free Predictive Dynamics	Fractals and hierarchical self-similarity of predictive computations across temporal scales.	LO+GO
29	Universal Error Coding	Common computational principles for error generation irrespective of receptive field complexity.	LO+GO
30	Canonical Microcircuit Generalization	Universal deployment of the canonical cortical microcircuit for inference across areas.	LO+GO
31	Cortico-Subcortical Homology	Shared predictive coding architecture between neocortex and basal ganglia / cerebellum.	LO+GO
32	Neuromodulatory Invariance	Consistent cholinergic/noradrenergic precision tuning across hierarchical stages.	LO+GO
33	Temporal Stability of Ubiquity	The consistent and non-transient presence of these mechanisms across the hierarchy over long recording sessions.	LO+GO
34	Hierarchical Order Stability	Evidence that the predictive mechanism is ubiquitous and not localized to a single hierarchical pole.	LO+GO
35	Population-Wide Ubiquity	Consistency of the mechanism across diverse neural populations and cell types throughout the hierarchy.	LO+GO
36	State-Independent Ubiquity	Presence of the mechanism across different brain states (e.g., wakefulness, sleep, anesthesia) throughout the hierarchy.	LO+GO

Supplementary Table S3: The factors of Hypothesis 3 (Ubiquity).

~~**Term/Context Description Definition (notation)** Oddball A surprising event, often a mismatch $= \{\mathrm{B}\} given = \{A\}$ Local Oddball (LO) An oddball that is immediate or short-term, often confounded by low-level adaptation mechanisms. $= \{\mathrm{AB}\} given = \{AA\}$ Global Oddball (GO) An oddball that is not LO, often contextual, that is not caused by adaptation mechanisms. $= \{\mathrm{AA}\} given = \{AB\}$~~

Term / Context	Description	Definition (notation)
Oddball	A surprising event, often a mismatch	$E = \{B\}$ given $S = \{A\}$
Local Oddball (LO)	An oddball that is immediate or short-term, often confounded by low-level adaptation mechanisms.	$E = \{AB\}$ given $S = \{AA\}$
Global Oddball (GO)	An oddball that is not LO, often contextual, that is not caused by adaptation mechanisms.	$E = \{AA\}$ given $S = \{AB\}$

Supplementary Table S4: Context definitions and formal expectation notations.

~~**# Study Identifier Modality / Preparation Inclusion Basis Bibliographic Citation** 1 Attinger et al., 2017 Mouse V1 (2-Photon Ca^{2+}) Comparative Set (Visuomotor Mismatch) *Cell* 169, 1291–1302 2 Bakhtiari et al., 2021 In Silico (Deep Neural Nets) Comparative Set (Hierarchical Streams) *Nat. Commun.* 12, 25164 3 Bastos et al., 2012 In Silico / Theory (Microcircuit) Comparative Set (Canonical Microcircuit) *Neuron* 74, 695–708 4 Bastos et al., 2020 Primate V1/V4/PFC (Laminar ECoG) Comparative Set (Laminar Rhythms) *Neuron* 108, 124–137 5 Bekinschtein et al., 2009 Human (Scalp EEG) Comparative Set (LO vs GO Originator) *PNAS* 106, 1672–1677 6 Chao et al., 2018 Primate Temporal/Frontal (ECoG) Comparative Set (Large-Scale Hierarchy) *Nat. Commun.* 9, 3188 7 Friston et al., 2010 Theory (Free Energy Principle) Canonical Extension (Bayesian Inference) *Nat. Rev. Neurosci.* 11, 127 8 Furutachi et al., 2024 Mouse V1 (2-Photon / Opto) Comparative Set (VIP/SST Disinhibition) *Nature* 633, 398–406 9 Garrett et al., 2020 Mouse V1 (2-Photon Ca^{2+}) Comparative Set (Novelty vs Deviance) *eLife* 9, e50340 10 Greedy et al., 2022 In Silico (Predictive Coding SNN) Comparative Set (Error Backpropagation) *NeurIPS* 35, 24213–24225 11 Hertag et al., 2020 In Silico (Spiking Interneurons) Comparative Set (SST/PV Dynamics) *eLife* 9, e57541 12 Jiang et al., 2024 In Silico (Energy-Based Model) Comparative Set (Sequence Prediction) *PLOS Comput. Biol.* 20, e1011801 13 Keller et al., 2012 Mouse V1 (2-Photon Ca^{2+}) Comparative Set (Sensorimotor Mismatch) *Neuron* 74, 809–815 14 Keller et al., 2018 Synthesis (Cortical Circuits) Comparative Set (Circuit Review) *Neuron* 100, 424–435 15 Kiebel et al., 2008 In Silico (Hierarchical Dynamics) Comparative Set (Temporal Hierarchy) *PLoS Comput. Biol.* 4, e1000209 16 Lao et al., 2023 Human (Auditory Scalp EEG) Comparative Set (Omission Response) *Sci. Adv.* 9, eabq8657 17 Lee et al., 2025 In Silico (Multi-Area Network) Comparative Set (Primate Oscillations) *PLOS Comput. Biol.* 21, e1013469 18 Mikulasch et al., 2023 In Silico / Theory (Plasticity) Comparative Set (Dendritic Error Review) *Trends Neurosci.* 46, 45–59 19 Nejad et al., 2025 Primate / In Silico (Laminar Model) Comparative Set (Layered Routing) *J. Neurosci.* 45, 1234 20 Payeur et al., 2021 In Silico (Dendritic Plasticity) Comparative Set (Burst Multiplexing) *Nat. Neurosci.* 24, 1010–1019 21 Rao et al., 2024 Synthesis (Sensorimotor Neocortex) Canonical Extension (25-Year Synthesis) *Nat. Neurosci.* 27, 1221–1235 22 Rao & Ballard, 1999 In Silico (Hierarchical Model) Canonical Extension (Foundational Model) *Nat. Neurosci.* 2, 79–87 23 Sacramento et al., 2018 In Silico (Dendritic Network) Comparative Set (Somatodendritic Model) *NeurIPS* 31, 8721–8731 24 Spratling et al., 2008 In Silico (Biased Competition) Comparative Set (PC-BC Synthesis) *Biol. Cybern.* 98, 283–292 25 Spratling et al., 2010 In Silico (V1 Microcircuit) Comparative Set (V1 Microcircuit Model) *J. Neurosci.* 30,~~

~~3531–3543 26 Srinivasan et al., 1982 Fly Retina (Intracellular) Canonical Extension (Redundancy Baseline) *Proc. R. Soc. Lond. B* 216, 427 27 VanDerveer et al., 2021 Human (Clinical EEG Cohort) Comparative Set (Sensory Gating Deficits) *Schizophr. Bull.* 47, 1385 28 Wacongne et al., 2011 Human (Auditory MEG) Comparative Set (Hierarchical Rule MMN) *PLoS Biol.* 9, e1001189 29 Wacongne et al., 2012 Human EEG & Spiking Network Comparative Set (Auditory Oddball Model) *J. Neurosci.* 32, 3665–3678 30 Westerberg & Xiong, 2025 Primate V1/V4/IT (Laminar MUA) Primary Index Study (MaDeLaNe) *bioRxiv* 10.1101/2024.10.02 31 Yamins et al., 2014 In Silico / Primate IT (HCNN) Comparative Set (Feedforward Baseline) *PNAS* 111, 8619–8624~~

#	Study Identifier	Modality / Preparation	Inclusion Basis	Bibliographic Citation
1	Attinger et al., 2017	Mouse V1 (2-Photon Ca^{2+})	Comparative Set (Visuomotor Mismatch)	*Cell* 169, 1291–1302
2	Bakhtiari et al., 2021	In Silico (Deep Neural Nets)	Comparative Set (Hierarchical Streams)	*Nat. Commun.* 12, 25164
3	Bastos et al., 2012	In Silico / Theory (Microcircuit)	Comparative Set (Canonical Microcircuit)	*Neuron* 74, 695–708
4	Bastos et al., 2020	Primate V1/V4/PFC (Laminar ECoG)	Comparative Set (Laminar Rhythms)	*Neuron* 108, 124–137
5	Bekinschtein et al., 2009	Human (Scalp EEG)	Comparative Set (LO vs GO Originator)	*PNAS* 106, 1672–1677
6	Chao et al., 2018	Primate Temporal/Frontal (ECoG)	Comparative Set (Large-Scale Hierarchy)	*Nat. Commun.* 9, 3188
7	Friston et al., 2010	Theory (Free Energy Principle)	Canonical Extension (Bayesian Inference)	*Nat. Rev. Neurosci.* 11, 127
8	Furutachi et al., 2024	Mouse V1 (2-Photon / Opto)	Comparative Set (VIP/SST Disinhibition)	*Nature* 633, 398–406
9	Garrett et al., 2020	Mouse V1 (2-Photon Ca^{2+})	Comparative Set (Novelty vs Deviance)	*eLife* 9, e50340
10	Greedy et al., 2022	In Silico (Predictive Coding SNN)	Comparative Set (Error Backpropagation)	*NeurIPS* 35, 24213–24225
11	Hertag et al., 2020	In Silico (Spiking Interneurons)	Comparative Set (SST/PV Dynamics)	*eLife* 9, e57541
12	Jiang et al., 2024	In Silico (Energy-Based Model)	Comparative Set (Sequence Prediction)	*PLOS Comput. Biol.* 20, e1011801
13	Keller et al., 2012	Mouse V1 (2-Photon Ca^{2+})	Comparative Set (Sensorimotor Mismatch)	*Neuron* 74, 809–815
14	Keller et al., 2018	Synthesis (Cortical Circuits)	Comparative Set (Circuit Review)	*Neuron* 100, 424–435
15	Kiebel et al., 2008	In Silico (Hierarchical Dynamics)	Comparative Set (Temporal Hierarchy)	*PLoS Comput. Biol.* 4, e1000209
16	Lao et al., 2023	Human (Auditory Scalp EEG)	Comparative Set (Omission Response)	*Sci. Adv.* 9, eabq8657
17	Lee et al., 2025	In Silico (Multi-Area Network)	Comparative Set (Primate Oscillations)	*PLOS Comput. Biol.* 21, e1013469
18	Mikulasch et al., 2023	In Silico / Theory (Plasticity)	Comparative Set (Dendritic Error Review)	*Trends Neurosci.* 46, 45–59
19	Nejad et al., 2025	Primate / In Silico (Laminar Model)	Comparative Set (Layered Routing)	*J. Neurosci.* 45, 1234
20	Payeur et al., 2021	In Silico (Dendritic Plasticity)	Comparative Set (Burst Multiplexing)	*Nat. Neurosci.* 24, 1010–1019
21	Rao et al., 2024	Synthesis (Sensorimotor Neocortex)	Canonical Extension (25-Year Synthesis)	*Nat. Neurosci.* 27, 1221–1235
22	Rao & Ballard, 1999	In Silico (Hierarchical Model)	Canonical Extension (Foundational Model)	*Nat. Neurosci.* 2, 79–87
23	Sacramento et al., 2018	In Silico (Dendritic Network)	Comparative Set (Somatodendritic Model)	*NeurIPS* 31, 8721–8731
24	Spratling et al., 2008	In Silico (Biased Competition)	Comparative Set (PC-BC Synthesis)	*Biol. Cybern.* 98, 283–292
25	Spratling et al., 2010	In Silico (V1 Microcircuit)	Comparative Set (V1 Microcircuit Model)	*J. Neurosci.* 30, 3531–3543
26	Srinivasan et al., 1982	Fly Retina (Intracellular)	Canonical Extension (Redundancy Baseline)	*Proc. R. Soc. Lond. B* 216, 427
27	VanDerveer et al., 2021	Human (Clinical EEG Cohort)	Comparative Set (Sensory Gating Deficits)	*Schizophr. Bull.* 47, 1385
28	Wacongne et al., 2011	Human (Auditory MEG)	Comparative Set (Hierarchical Rule MMN)	*PLoS Biol.* 9, e1001189
29	Wacongne et al., 2012	Human EEG & Spiking Network	Comparative Set (Auditory Oddball Model)	*J. Neurosci.* 32, 3665–3678
30	Westerberg & Xiong, 2025	Primate V1/V4/IT (Laminar MUA)	Primary Index Study (MaDeLaNe)	*bioRxiv* 10.1101/2024.10.02
31	Yamins et al., 2014	In Silico / Primate IT (HCNN)	Comparative Set (Feedforward Baseline)	*PNAS* 111, 8619–8624

Supplementary Table S5: 31-Paper Literature Corpus Registry with modality, inclusion basis, and provenance.

~~**Model Name Parameters Architecture & Provenance** gemma-3-27b-it 27B~~

~~Google DeepMind (Dense Transformer) gemma-4-31b-it 31B Google DeepMind (Extended Reasoning Architecture) deepseek-r1-distill-llama-70b 70B DeepSeek-AI / Meta Llama Distillation gpt-oss-claude-4.5-sonnet 20B Community Distillation / Reasoning mistral-nemo-12b-thinking 12B Mistral AI & NVIDIA olmo-3-32b-think 32B Allen Institute for AI (Ai2) phi-4-reasoning-plus 14B Microsoft Research qwen3-14b-gemini-3-pro 14B Alibaba Cloud / Community Fine-tune qwen3-14b-claude-4.5-sonnet 14B Alibaba Cloud / Community Fine-tune qwen3.5-40b-claude-4.5-opus 40B Alibaba Cloud / High-Capacity MoE~~

Model Name	Parameters	Architecture & Provenance
gemma-3-27b-it	27B	Google DeepMind (Dense Transformer)
gemma-4-31b-it	31B	Google DeepMind (Extended Reasoning Architecture)
deepseek-r1-distill-llama-70b	70B	DeepSeek-AI / Meta Llama Distillation
gpt-oss-claude-4.5-sonnet	20B	Community Distillation / Reasoning
mistral-nemo-12b-thinking	12B	Mistral AI & NVIDIA
olmo-3-32b-think	32B	Allen Institute for AI (Ai2)
phi-4-reasoning-plus	14B	Microsoft Research
qwen3-14b-gemini-3-pro	14B	Alibaba Cloud / Community Fine-tune
qwen3-14b-claude-4.5-sonnet	14B	Alibaba Cloud / Community Fine-tune
qwen3.5-40b-claude-4.5-opus	40B	Alibaba Cloud / High-Capacity MoE

Supplementary Table S6: Open-weight LLM evaluator models evaluated in this work.

~~**Evaluator Comparison Paired N Pearson r MAE Directional Concordance** Human–Human (H_1 vs H_2) 299 0.390 0.308 88.96% Human Consensus vs LLM Council 175 0.162 0.301 93.14% Council Inter-Model Average 584 0.540 0.284 85.00% Shuffled Evidence (Null H_0) 191 -0.087 0.662 56.54%~~

Evaluator Comparison	Level	Paired N	Pearson r	p	MAE	Concordance
Human–Human (H_1 vs H_2)	Factor cell	299	0.390	< 0.0001	0.308	88.96%
Human Consensus vs LLM Council	Study level	25	0.529	0.0125	—	—
Human Consensus vs LLM Council	Factor cell	175	0.162	0.0322	0.301	93.14%
Council Inter-Model Average	Factor cell	584	0.540	< 0.0001	0.284	85.00%
Shuffled Evidence (Null H_0)	Factor cell	191	-0.087	0.2314	0.662	56.54%

Supplementary Table S7: Evaluator Strata Agreement and Error Hierarchy Summary ($K = 2$ independent human domain experts). The independent study-level comparison ($N = 25$ papers with jointly evaluated factor ratings; Pearson $r = 0.529$, $p = 0.0125$, 95% CI [0.169, 0.764]) treats the independent paper as the unit of analysis. Factor-cell comparisons are descriptive summaries across scored cells nested within papers. Directional concordance denotes agreement in sign on non-zero factor cells.

~~**Adjudication Class Cells (N) Human Supported Council Supported Primary Failure / Divergence Mode** Theoretical Model / Synthesis 15 100.0% (15/15) 86.7% (13/15) Council magnitude attenuation on broad claims Direct Empirical Electrophysiology 7 100.0% (7/7) 71.4% (5/7) Subtle cell-type specificity (SST vs PV gain) Hierarchical / Multi-Area Studies 5 80.0% (4/5) 60.0% (3/5) Conflation of local vs global deviance scope **Total Adjudicated Disagreements 27 96.3% (26/27) 77.8% (21/27) Passage-grounded technical concordance**~~

Adjudication Category	**Cells (N)**	**Proportion**	**Primary Disagreement / Failure Mode**
Directly supported	6	20.0% (6/30)	Full primary text, equation, or figure corroboration of Council assignment
Partially supported	18	60.0% (18/30)	Relevant primary evidence present; conservative magnitude attenuation or specificity nuance
Unsupported attribution	6	20.0% (6/30)	Subtype gain conflation ($N = 3$), laminar error inversion ($N = 1$), untested modality penalty ($N = 1$), cross-paper attribution ($N = 1$)
Total Audited Disagreements	**30**	**100.0% (30/30)**	**Diagnostic audit of $\lvert H - C\rvert \geq 0.50$ cells ($N = 296$ eligible paired cells)**

Supplementary Table S8: Diagnostic ~~Passage-Support Adjudication across 27 Prespecified Disagreement Cells .~~ Primary-Source Audit across 30 Numerically Selected Large Human–Council Disagreement Cells ($\lvert H - C\rvert \geq 0.50$ among 296 eligible paired cells; 10.1% of paired universe). Evaluates whether large disagreements reflect unsupported evidence attribution or differences in magnitude, contextual interpretation, or biological specificity despite relevant primary-source evidence. Because this diagnostic sample was enriched for large discrepancies, these proportions characterize disagreement modes and do not represent corpus-wide accuracy or hallucination rates.

Decoding Setting / Comparison	**Evaluated Units (N)**	**Correlation**	**MAE**	**Paper Ranking Stability**
Temperature Stability (`gemma-4-31b-it`)				
$T = 0.00$ vs. $T = 0.35$	679 factor cells	Pearson $r = 0.9900$	0.0266	Spearman $\rho_{\text{rank}} = 0.9596$
$T = 0.00$ vs. $T = 0.70$	679 factor cells	Pearson $r = 0.9760$	0.0421	Spearman $\rho_{\text{rank}} = 0.9212$
$T = 0.35$ vs. $T = 0.70$	679 factor cells	Pearson $r = 0.9816$	0.0367	Spearman $\rho_{\text{rank}} = 0.9482$
Replicate Reliability (`gemma-4-31b-it`)				
Replicates at $T = 0.00$	639 triples	Mean $\bar{r} = 0.9905$	0.0126	Spearman $\rho_{\text{rank}} \geq 0.9818$
Replicates at $T = 0.35$	619 triples	Mean $\bar{r} = 0.9724$	0.0384	Spearman $\rho_{\text{rank}} \geq 0.9782$
Replicates at $T = 0.70$	572 triples	Mean $\bar{r} = 0.9444$	0.0585	Spearman $\rho_{\text{rank}} \geq 0.9745$

~~**Experimental Condition**~~**Supplementary Table S9:** ~~**Calls (N) Mean Score ±SD Displacement $\Delta_{\text{LO-GO}}$ Paper Ranking Stability** $T = 0.00$ (Deterministic)~~ Factorial Robustness Analysis across Sampling Temperatures and Stochastic Replicates for the Complete Evaluation Model (`gemma-4-31b-it`, 279~~0.463 ± 0.381 +0.041 Spearman $\rho_{\text{rank}} = 1.000$ (Baseline) $T = 0.35$ (Moderate)~~ /279 ~~0.458 ± 0.384 +0.039 Spearman $\rho_{\text{rank}} = 0.9841$ $T = 0.70$ (Exploratory)279 0.451 ± 0.392 +0.036 Spearman $\rho_{\text{rank}} = 0.9677$ Repeat 1 279 0.460 ± 0.382 +0.040 Spearman $\rho_{\text{rank}} = 0.9818$ Repeat 2 279 0.457 ± 0.386 +0.038 Spearman $\rho_{\text{rank}} = 0.9782$ Repeat~~ factorial conditions). Quantitative factorial statistics are reported exclusively for `gemma-4-31b-it` because it completed full factorial coverage across all 31 papers × 3 temperatures × 3 repeats. Extended chain-of-thought models (`phi-4-reasoning-plus` and `olmo-3-32b-think`) completed 130/279 ~~0.455 ± 0.389 +0.038 Spearman $\rho_{\text{rank}} = 0.9745$ Westerberg & Xiong (2025) — Divergent Outlier $\Delta = -0.400$ Rank 28–29/31 across all T conditions Leave-One-Model-Out (LOMO)10 folds $\Delta_{\text{Total}} \in [-0.105, -0.093]$ Negative throughout H2 $\Delta \in [-0.228, -0.208]$~~ and 30/279 calls, respectively, but frequently exhausted dynamic generation token budgets during chain-of-thought reasoning prior to emitting the terminal JSON scoring block, precluding equivalent comparative factorial statistics. Paper ranking stability measures Spearman correlation between paper mean scores across conditions ($N = 29$ evaluable papers).

~~Supplementary Table S9~~**Panel A: Paired Score Sensitivity to Identity Removal and Evidence Restriction** ~~Factorial Robustness Analysis across 837 LLM inference calls including Paper Ranking and LOMO Sensitivity .~~

~~**Evidence Manipulation Calls (N) Mean LO Score Mean GO Score Displacement $\Delta_{\text{LO-GO}}$** Intact Evidence (FULL) 18 0.512 ± 0.081 0.474 ± 0.076 +0.038 (LO > GO) Shuffled Evidence (H_0)18 0.489 ± 0.092 0.493 ± 0.088 −0.004 (Neutral)Prior Only (Zero Evidence) 18 0.450 ± 0.110 0.450 ± 0.108 +0.000 (Null)~~

Comparison	Paired Cells (N)	Pearson r	MAE	Scientific Interpreta
FULL vs MASKED Text	112	0.9937	0.0089	Tests deper explicit pub identity
FULL vs ABSTRACT Only	88	0.7274	0.0758	Tests sensit restriction c paper evide

Panel B: Abstract-Only Sensitivity by Hypothesis Family

Hypothesis Family	Paired Cells (N)	Pearson r	MAE	Coverage Note
Hypothesis 1 (Predictive Suppression)	30	0.6003	0.1444	5 papers with evaluable abstract evidence
Hypothesis 2 (Feedforward Error)	38	0.6776	0.0614	4 papers with evaluable abstract evidence
Hypothesis 3 (Ubiquity)	20	1.0000	0.0000	2 papers; insufficient coverage for general inference

Panel C: Zero-Manuscript-Evidence Control (Prior Only)

Input Condition	Calls (N)	LO Mean [±SD]	GO Mean [±SD]	Context Contrast ($\Delta_{\text{LO-GO}}$)
Intact Evidence (FULL Reference)	18	0.512 ± 0.081	0.474 ± 0.076	+0.038 (Active context differentiation)
Prior Only (Zero Manuscript Evidence)	18	0.450 ± 0.110	0.450 ± 0.108	0.000 (Parity across contexts)

Supplementary Table S10: ~~Evidence Grounding vs. Prior Bias Sensitivity Analysis (54 Calls).~~ Evidence Availability, Publication-Identity, and Prior-Only Sensitivity Analyses. De-identifying full manuscript text produced minimal score adjustments (FULL vs MASKED, Panel A), whereas restricting available evidence to titles and abstracts produced larger deviations (FULL vs ABSTRACT, Panel A) with the largest attenuation in predictive suppression mechanisms (Panel B). In the zero-manuscript-evidence condition (Panel C), models retained baseline affirmative support from the ontology and task framing, but the Local vs. Global Oddball context differentiation observed under intact manuscript evidence was completely absent. Abstract-only Hypothesis 3 coverage was limited to 20 paired cells from two papers and is therefore not interpreted as evidence of general invariance across the literature.

~~**Ablation Condition Paired Cells (N) Pearson r MAE Displacement $\Delta_{\text{LO-GO}}$**~~ ~~FULL Baseline (With Figures) 533 1.000 0.000 +0.0408 NO_FIGURE (Figures Stripped) 533 0.8539 0.2253 +0.0565~~

Ablation Condition	Paired Cells (N)	Pearson r	MAE	Directional Concordance
FULL Baseline (With Figures)	533	1.000	0.000	100.0% (Baseline)
NO_FIGURE (Figures Stripped)	533	0.8539	0.2253	99.42% (518/521 non-zero cells)

Supplementary Table S11: Figure-Text-Removal Sensitivity Summary ($N = 31$ papers).

~~**Dimension Stratum Papers (N) Council $\Delta_{\text{LO-GO}}$ Directional Trend**~~ ~~Study~~

~~Type Empirical 13 −0.009 GO > LO Study Type Computational / Theory 18 −0.013 GO > LO Modality In Silico 19 −0.013 GO > LO Modality Human In Vivo 7 −0.059 GO > LO Modality Animal In Vivo 5 +0.087 LO > GO Publication Era 1982–2014 12 −0.024 GO > LO Publication Era 2015–2020 7 −0.041 GO > LO Publication Era 2021–2025 12 +0.038 LO > GO Publication Status Peer-Reviewed Journal 30 −0.010 GO > LO Publication Status Preprint (Westerberg & Xiong 2025) 1 −0.400 GO ≫ LO~~

Dimension	Stratum	Papers (N)	Council LO Mean	Council GO Mean
Study Type	Empirical	13	0.419	0.428
Study Type	Computational / Theory	18	0.351	0.365
Modality	In Silico	19	0.363	0.376
Modality	Human In Vivo	7	0.357	0.416
Modality	Animal In Vivo	5	0.511	0.424
Publication Era	1982–2014	12	0.368	0.393
Publication Era	2015–2020	7	0.447	0.488
Publication Era	2021–2025	12	0.347	0.309
Publication Status	Peer-Reviewed Journal	30	0.381	0.392
Publication Status	Preprint (Westerberg & Xiong 2025)	1	0.166	−0.244

Supplementary Table S12: Corpus Subgroup Sensitivity Analysis across Taxonomic Strata. Demonstrates that council-estimated evidence support across Local and Global Oddball contexts varies by experimental modality and publication era, with Westerberg & Xiong (2025) exhibiting a distinct negative Global Oddball profile.

~~**Validation Axis Evaluated Figures (N) Correct Partial / Minor Error Overall Accuracy Rate** Axis A: Numerical Trend & Data Extraction 10 9 1 (Minor plot type phrasing) 100.0% PASS (10/10) Axis B: Empirical Effect Direction 10 10 0 100.0% Correct (10/10) Axis C: HPC-36 Ontology Factor Mapping 10 9 1 (Partial invertebrate mapping) 100.0% PASS (10/10) **Overall Manual Validation Outcome 10 Figures 10/10 PASS 0 Fatal Inversions 100.0% PASS**~~

Validation Axis	Evaluated (N)	Strictly Correct	Partial / Minor	Audit Pass Rate
Axis A: Numerical Trend & Data Extraction	10	9	1 (Minor phrasing)	100.0% PASS (10/10)
Axis B: Empirical Effect Direction	10	10	0	100.0% PASS (10/10)
Axis C: HPC-36 Ontology Factor Mapping	10	9	1 (Invertebrate)	100.0% PASS (10/10)
Overall Manual Validation Outcome	**10 Figures**	**10/10**	**0 Fatal Inversions**	**100.0% PASS**

Supplementary Table S13: Manual Expert Validation Summary of Generated Multimodal Figure Descriptions (N = 10 ~~prespecified~~ representative figures audited against ~~primary-source PDFs)~~primary source PDFs). All 10 figures met the prespecified audit criterion (Audit Pass Rate: 100.0%); minor phrasing and partial ontology mappings are reported separately.

Context	Hyp.	N	Human Mean	Council Mean	Human−Council [95% CI]	$t(df), p$
Local Oddball (LO)	H_1	21	$+0.537 \pm 0.041$	$+0.337 \pm 0.027$	$+0.200$ $[0.111, 0.289]$	$t(20) = 4.69, p = 0.0001$
	H_2	23	$+0.552 \pm 0.045$	$+0.363 \pm 0.026$	$+0.189$ $[0.117, 0.261]$	$t(22) = 5.42, p < 0.0001$
	H_3	14	$+0.520 \pm 0.034$	$+0.268 \pm 0.045$	$+0.252$ $[0.161, 0.344]$	$t(13) = 5.95, p < 0.0001$
Global Oddball (GO)	H_1	18	$+0.600 \pm 0.048$	$+0.299 \pm 0.031$	$+0.302$ $[0.182, 0.422]$	$t(17) = 5.32, p < 0.0001$
	H_2	22	$+0.560 \pm 0.043$	$+0.259 \pm 0.049$	$+0.300$ $[0.205, 0.396]$	$t(21) = 6.54, p < 0.0001$
	H_3	11	$+0.557 \pm 0.065$	$+0.312 \pm 0.057$	$+0.244$ $[0.080, 0.408]$	$t(10) = 3.32, p = 0.0077$

Supplementary Table S14: Primary Paper-Level Hypothesis Calibration and Concordance ($K = 2$ Human Consensus vs. Autonomous Council). Analysis is restricted to independent papers where both human experts ($K = 2$) evaluated mechanisms within the specific hypothesis. Council models were averaged within paper first. SEM denotes standard error across independent papers (SD/$\sqrt{N}$). Paired differences (Human−Council mean difference) evaluate systematic score attenuation. Paired t-tests ($t(df)$) and two-sided p-values evaluate the difference in mean support.

Omitted Evaluator Model	Paired N	Pearson r	p	Spearman ρ	p	MAE
`deepseek-r1-distill-llama-70b`	1710	0.9553	< 0.0001	0.9589	< 0.0001	0.0465
`gemma-3-27b`	1733	0.9747	< 0.0001	0.9472	< 0.0001	0.0408
`gpt-oss-claude-4.5-sonnet`	1801	0.9920	< 0.0001	0.9844	< 0.0001	0.0086
`mistral-nemo-12b-thinking`	1775	0.9732	< 0.0001	0.9698	< 0.0001	0.0239
`olmo-3-32b-think`	1798	0.9900	< 0.0001	0.9800	< 0.0001	0.0122
`phi-4-reasoning-plus`	1778	0.9825	< 0.0001	0.9735	< 0.0001	0.0255
`qwen3-14b-gemini-3-pro`	1800	0.9797	< 0.0001	0.9649	< 0.0001	0.0243
`qwen3.5-40b-claude-4.5-opus`	1724	0.9497	< 0.0001	0.9178	< 0.0001	0.0440
Evaluator profiles with hypothesis-level scores but unassigned individual factor cells:						
`gemma-4-31b`	1816	—	—	—	—	—
`gpt-oss-safeguard-120b`	1816	—	—	—	—	—

Supplementary Table S15: Leave-One-Model-Out Council Score Stability Analysis across Evaluator Architectures. Evaluates the sensitivity of aggregate Council scores to single-model removal by comparing the complete Council continuous mean against the Council recomputed after omission of each contributing evaluator across matched factor cells ($N = 1710$–1801). Across all active evaluator omissions ($N = 8$ model omissions), Council scores remained highly stable: mean Pearson $r = 0.9747$ (95% CI: $[0.9619, 0.9874]$, range 0.9497–0.9920), mean Spearman $\rho = 0.9621$ (95% CI: $[0.9441, 0.9800]$, range 0.9178–0.9844), and mean MAE = 0.0282 (95% CI: $[0.0163, 0.0402]$, range 0.0086–0.0465). Confidence intervals are computed using Student's t-distribution ($df = N_{\text{omissions}} - 1 = 7$) with model omission as the unit. MAE is reported in original $[-1.0, +1.0]$ score units. Two evaluator profiles contributed hypothesis-level Council scores but no individual factor-cell scores and therefore were not evaluable in this factor-level leave-one-out analysis.